

MODELLING SPATIAL SURVIVAL WITH EXTERNAL COVARIATES

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Summary: This paper presents a general approach for analyzing spatial survival in the plane in the presence of external time-dependent and time-independent covariates. A small normally oriented interval is moving along a fixed route in the plane, crossing points of the planar Poisson process (points with non-moving coordinates). Each crossing leads to termination of the process with a given probability, which can depend on time and on external covariates. These covariates are modelled by deterministic functions and by stochastic processes as well. The probability of passing the route without termination is derived. A safety at sea application is discussed.

1. Introduction

In our previous paper (Finkelstein, 1998) a model for spatial survival was developed and applied to the safety at sea assessment. It was based on deriving the equivalent failure rate and reducing a planar setting to a one-dimensional case. In the current study we mostly deal with external stochastic covariate factors, which lead to random failure rates. The weather influence (time-dependent factor) or different uncertainties in the quality of a ship's navigation or motion control systems (time-independent factors) are the relevant examples. In what follows in this section, we describe the general

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setting of the problem and briefly cover the results of Finkelstein (1998), necessary for the subsequent presentation.

Consider a system subject to a stochastic point process of shocks with rate $h(t)$. We interpret the word shock in a very broad sense as some point, possibly harmful, influence on a system (Barlow and Proschan, 1975). Each shock can lead with a given probability to a fatal failure of a system, resulting in termination of the process, and this will be called an "accident". The probability of the system performing without accidents in the time interval $(0, t]$ is of interest. One can effectively deal with this problem only under the assumption that the point process is Poisson and this will be assumed in what follows.

Assume now that a shock, occurring in $(t, t + dt]$, independently of the previous shocks, leads to an accident with probability $\theta(t)$, and does not cause any changes in the system with complementary probability $1 - \theta(t)$. Denote by T_a a random time to an accident and by $F_a(t) = \Pr\{T_a \leq t\}$ the corresponding distribution function (DF). It follows, for instance, from Block et al. (1985) that

$$P(t) = 1 - F_a(t) = \exp \left\{ - \int_0^t \theta(x) h(x) dx \right\}, \quad (1)$$

which means that $\lambda_a(t) = \theta(t) h(t)$ is a failure rate for $F_a(t)$.

In Finkelstein (1998) this one-dimensional setting was applied to the plane and specifically for assessing the ship's safe (without accident) passing of the route. Denote by $\{N(B)\}$ the nonhomogeneous Poisson point process in the plane, where $N(B)$ is the corresponding random measure: i.e. the random number of points in $B \subset \mathfrak{R}^2$, where a domain B belongs to the Borel σ -algebra in $\mathfrak{R}^2$. We shall consider points as prospective point influences on

our system (unknown shallows for the ship, for instance). Denote the rate of this process by $h(\xi)$. Denote by R_{ξ_1, ξ_2} a fixed continuous curve connecting ξ_1 and ξ_2 – two distinct points in the plane to be called a route. A point (a ship in our application) is moving in one direction along the route. Every time it “crosses a point” (see later) of the process $\{N(B)\}$ an accident can happen with a given probability. We are interested in assessing the probability of moving along R_{ξ_1, ξ_2} without accident. Let r be the distance from ξ_1 to the current point on the route (coordinate) and $h(r)$ denote the rate in $(r, r + dr]$. For defining the corresponding Poisson measure the dimensions of objects under consideration should be taken into account.

Let $\bar{\gamma}_n(r) \equiv (\gamma_n^+(r), \gamma_n^-(r))$ be a small and “symmetrical” (for simplicity) interval of length $\bar{\gamma}_n(r) = 2\gamma_n^+(r) = 2\gamma_n^-(r)$ on a normal to R_{ξ_1, ξ_2} at the point with coordinate r , where upper indexes denote the corresponding direction: $\gamma_n^+(r)$ is on one side of R_{ξ_1, ξ_2} while $\gamma_n^-(r)$ is on the other. Let $\bar{R}$ be the length of R_{ξ_1, ξ_2} : $\bar{R} \equiv |R_{\xi_1, \xi_2}|$. The interval $\bar{\gamma}_n(r)$ is moving along R_{ξ_1, ξ_2} , crossing points of the random field. Now we can easily define the corresponding *equivalent rate* of occurrence of points, $h_e(r)$:

$$h_e(r) = \lim_{\Delta r \rightarrow 0} \frac{E[N(B(r, \Delta r, \gamma_n(r)))]}{\Delta r}, \quad (2)$$

where $N(B(r, \Delta r, \gamma_n(r)))$ is the random number of points crossed by the interval $\bar{\gamma}_n(r)$, moving from r to $r + \Delta r$.

It can be easily seen (Finkelstein, 1998), that for $\Delta r \rightarrow 0$ and $\gamma_n(r)$ sufficiently small, and assuming that $h(\xi)$ is continuous:

$$\begin{aligned} E[N(B(r, \Delta r, \gamma_n(r)))] &= \int_{B(r, \Delta r, \gamma_n(r))} h(\xi) dS(\delta(\xi)) \\ &\stackrel{\Delta r \rightarrow 0}{\approx} \gamma_n(r) h(r) \Delta r [1 + o(1)], \end{aligned} \quad (3)$$

where $\delta(\xi)$ is a neighbourhood of ξ and $S(\delta(\xi))$ is its area. Equation (3) leads to the expression for the equivalent rate of the corresponding one-dimensional point process (which is obviously also nonhomogeneous Poisson):

$$h_e(r) = \gamma_n(r) h(r) [1 + o(1)]. \quad (4)$$

As previously, assume that crossing of a point with a coordinate r leads to an accident with probability $\theta(r)$ and to the "survival" with the complementary probability $\bar{\theta}(r) = 1 - \theta(r)$. Relations (2)–(4) actually define the one-dimensional parameterization of the problem and therefore the analog of formula (1) can be used. Denote by R a random distance from the initial point of the route ξ_1 till the point on the route where an accident occurs. Similarly to (1), the probability of passing the route R_{ξ_1, ξ_2} without accidents can be expressed as follows:

$$\Pr\{R > \bar{R}\} \equiv P(\bar{R}) = \exp\left\{-\int_0^{\bar{R}} \theta(r) h_e(r) dr\right\}. \quad (5)$$

Thus

$$\lambda_a(r) \equiv \theta(r) h_e(r) \quad (6)$$

is the failure rate of the $DF: F_a(r) \equiv 1 - P(r)$.

2. Stochastic intensity

Assume that $\lambda_a(r)$ in (6) is now random and is defined, for instance, as in Yashin and Manton (1997) by an unobserved covariate stochastic process $Y = Y_r, r \geq 0$. Denote the resulting stochastic process by $\lambda_a(r, Y_r)$. This process models an external stochastic influence on our object. The relevant examples will be considered below. Suppose that the following important

relation holds:

$$\Pr \{R > r \mid y(s), 0 \leq s \leq r\} = \exp \left\{ - \int_0^r \lambda_a(u, y(s); 0 \leq s \leq r) du \right\}, \quad (7)$$

where $y(s); 0 \leq s \leq r$ is the trajectory (history) of the process Y in $[0, r]$. This assumption means that the conditional failure rate exists and therefore the corresponding conditional DF is absolutely continuous. It is well known (see, e.g. Kebir, 1991) that in this case the following “unconditional” equation holds:

$$\Pr \{R > r\} \equiv P(r) = E \left[\exp \left\{ - \int_0^r \lambda_a(u, Y_r) du \right\} \right], \quad (8)$$

where the expectation is taken with respect to the stochastic process Y_r . We assume also that this expectation exists. In this case the stochastic failure rate is usually called the stochastic intensity or the hazard rate process (Finkelstein, 2000). Evaluating the right-hand side in (8) is usually difficult and can be performed explicitly only in simple specific cases as in Singpurwalla (1995). As follows from Yashin and Manton (1997), under assumption (7) and the following measurability condition:

$$E \left[\int_0^r \lambda_a(u, Y_r) du \right] < \infty$$

equation (8) can be continued as

$$P(r) = \exp \left\{ - \int_0^r E[\lambda_a(u, Y_r) \mid R > u] du \right\} = \exp \left\{ - \int_0^r \bar{\lambda}_a(u) du \right\}, \quad (9)$$

where $\bar{\lambda}_a(r)$ is the corresponding unconditional or observed failure rate, which describes the random variable R . Thus, the relation between the

observed and conditional failure rates is

$$\bar{\lambda}_a(r) = E[\lambda_a(r, Y_r) | R > r]. \quad (10)$$

It is clear that equations (9) and (10) constitute a reasonable tool for obtaining the probability of survival on the route $P(\bar{R})$, but as has been mentioned, the corresponding explicit derivations can be performed only in some specific simple cases. On the other hand, it can help in analyzing some important descriptive properties.

Consider the following example in the framework of our model. Assume a specific form of the corresponding hazard rate process. For instance, let the probability $\theta(r)$ be indexed by a parameter $Y : \theta(r, Y)$, which is interpreted as a non-negative random variable with support in $[0, \infty)$ and having probability density function $\pi(y)$. In the sea safety application this randomization can be due to the unknown characteristics of the navigation (or (and) collision avoidance) onboard system, for instance. There can be other interpretations as well. Thus, the specific case, when Y in relations (8)–(10) is a random variable, is considered. An important condition (7) is fulfilled, as the trajectory of Y for this setting means just a realized value y for which, in accordance with the previous section, the exponential formula (7) is valid. It turns out that the observed failure rate $\bar{\lambda}_a(r)$ is actually the mixture failure rate for this case (Lynn and Singpurwalla, 1997, Finkelstein and Esaulova, 2000a and 2000b):

$$\bar{\lambda}_a(r) = \int_0^{\infty} \lambda_a(r, y) \pi(y|r) dy, \quad (11)$$

where $\pi(y|r)$ is the conditional probability density function of Y given that $R > r$.

$$\pi(y|r) = \frac{\pi(y) P(r, y)}{\int_0^\infty P(r, y) \pi(y) dy} \quad (12)$$

and $P(r, y)$ is defined similarly to (5), where $\lambda_a(r)$ is substituted by $\lambda_a(r, y) = \theta(r, y) h_e(r)$ for fixed y .

Relations (11) and (12) constitute a reasonable tool for analyzing the shape of the observed failure rate $\bar{\lambda}_a(r)$ and this analysis is important for reliability and safety applications. As shown in Finkelstein and Esaulova (2000a), the shape of $\bar{\lambda}_a(r)$ can differ dramatically from the shape of the conditional failure rate $\lambda_a(r, y)$. Assume, for example, a specific multiplicative form of parameterization:

$$\theta(r, Y) h_e(r) = Y \theta(r) h_e(r). \quad (13)$$

It is well known that, if $\theta(r) h_e(r)$ is decreasing, then the observed failure rate defined via relation (13) is decreasing (e.g., Barlow, 1985). It turns out that even if $\theta(r) h_e(r)$ is sharply increasing, $\bar{\lambda}_a(r)$ can still decrease at least for sufficiently large r ! (see examples in Finkelstein and Esaulova, 2000a). Thus, the random parameter changes the aging properties of the corresponding distribution function.

It is worth noting that a similar randomization can be performed in the second multiplier $h_e(r) \approx y_n(r) h(r)$ in the right-hand side of (6), but this should be the subject of a special study.

For “highly reliable systems” in which, for instance, $h(r) \rightarrow 0$ uniformly in $r \in [0, \bar{R}]$, one can easily obtain the following obvious unconditional asymptotic relations from (8) and (10):

$$P(\bar{R}) = \left[1 - E \left[\int_0^{\bar{R}} \lambda_a(r, Y) dr \right] \right] [1 + o(1)] \quad (14)$$

$$\begin{aligned}
&= \left[1 - \int_0^{\bar{R}} E[\lambda_a(r, Y) dr] \right] [1 + o(1)] \\
&= \left[1 - \int_0^{\bar{R}} \bar{\lambda}_a(r) dr \right] [1 + o(1)] ,
\end{aligned}$$

where

$$\bar{\lambda}_a(r) = E[\lambda_a(r, Y) | R > r] = E[\lambda_a(r, Y)] [1 + o(1)] . \quad (15)$$

Eventually (compare with (11)):

$$\bar{\lambda}_a(r) = \int_0^{\infty} \lambda_a(r, y) \pi(y) dy [1 + o(1)] . \quad (16)$$

On the other hand, as the exponential function is convex and the expectation exists, we can apply the Jensen inequality to the right-hand side of (8) and obtain a simple lower bound for the general case:

$$P(\bar{R}) \geq \exp \left\{ E \left[- \int_0^{\bar{R}} \lambda_a(r, Y_r) dr \right] \right\} . \quad (17)$$

Example 1.

Let the failure rate in the right-hand side in the model (13) be constant for fixed y :

$$\theta(r) \equiv \theta; \quad h_e(r) \equiv h_e \quad (18)$$

and Y be exponentially distributed with parameter ϑ . Applying relation (11), it can easily be seen (Finkelstein and Esaulova, 2001a) that:

$$\bar{\lambda}_a(r) = \frac{\theta h_e}{r \theta h_e + \vartheta} \quad (19)$$

whereas using equation (16), an approximately constant failure rate is obtained for $h_e \rightarrow 0$ or $r \rightarrow 0$:

$$\bar{\lambda}_{af}(r) = \frac{\theta h_e}{\vartheta} [1 + o(1)] ,$$

which alternatively under the same conditions can be derived directly from (19). The lower bound (17) in this case is

$$P(r) \geq \exp \left\{ -\frac{\theta h_e}{\vartheta} r \right\}.$$

As it was already stated, the shape of $\bar{\lambda}_a(r)$ can differ dramatically from the shape of the conditional failure rate $\lambda_a(r, y)$. For this example the conditional failure rate is a constant while relation (19) implies that

$$\bar{\lambda}_a(r) = \frac{1}{r} [1 + o(1)] \text{ to } t \rightarrow \infty.$$

3. Time-dependent covariates

Up till now the external covariates were time-independent. It is clear that in the safety at sea application, for instance, there exist external influences, which are definitely time-dependent. One of the possible models of this kind was considered in Finkelstein (1998), where other ships played a role of moving obstacles and the intensity of navigation was interpreted as the time-dependent rate $h(t, \xi)$. In this section some other settings will be considered at a descriptive level. We also plan to present a more advanced mathematical treatment of the model elsewhere.

Let the following natural multiplicative model hold for the rate of the spatial-temporal (Cox and Isham, 1980) stochastic point process of obstacles:

$$h(t, \xi) = \varphi(t) h(\xi), \quad (20)$$

where $\varphi(t)$ is a deterministic function of time. As usual for proportional models, we can assume that $\varphi(t)$ describes the influence of environment and that $\varphi(t) \equiv 1$ for some baseline (reference) environment.

In the safety at sea application $\varphi(t)$ can describe, for instance, the weather influence or tidal effects in a region close to shore (the lower the level of the water the higher the probability of shallows), where the time t should be

counted from the start of the trip under consideration. As previously, the ship moves along the fixed route R_{ξ_1, ξ_2} and it starts at ξ_1 , $t = 0$. Assume for simplicity that the speed v of the ship is constant. Similarly to equations (2)–(4), and taking into account equation (20), the equivalent rate of crossing points of the field of obstacles is derived for this case in the following way:

$$E[N(B(r, \Delta r, \gamma_n(r)))] = \int_{B(r, \Delta r, \gamma_n(r))} h(t, \xi) dS(\delta(\xi)) \quad (21)$$

$$\stackrel{\Delta r \rightarrow 0}{=} \varphi\left(\frac{r}{v}\right) \gamma_n(r) h(r) \Delta r [1 + o(1)] ,$$

$$h_e(r) = \varphi\left(\frac{r}{v}\right) \gamma_n(r) h(r) [1 + o(1)] . \quad (22)$$

The quotient r/v is the time for the moving object to reach the coordinate r . Relation (21) can easily be generalized to the case of the non-constant speed $v(r)$:

$$\varphi(t) = \varphi\left(\int_0^r \frac{dr}{v(r)}\right)$$

Thus, everything is again parameterized by r and equations (5)–(6) can be used with evident modifications. It is worth noting that the rate of the initial point process $h(t, \xi)$ is time-dependent, while the equivalent failure rate $h_e(r)$, obtained in the described way, is not! The stochastic process φ_t , $t \geq 0$ with trajectories $\varphi(u)$ in $[0, t]$ can also be considered in this approach. The equivalent failure rate will then become the hazard rate process and the approach of the previous section can be used. It should be stated once more that explicit results deriving the probability of survival based on hazard rate processes modelling are obtainable only for very simple special cases.

Perhaps, from a practical point of view the most natural way of introducing the time-dependency in the model is to consider the time-dependent probability

$\theta(t, r)$. This notation now means that, if the ship has a coordinate r (defined for the given route R_{ξ_1, ξ_2}) at time t , then the probability of an accident is $\theta(t, r)$. It is clear that the weather influence, for instance, can be modelled in this way, as the probability of an accident directly depends on the weather conditions at the current moment in time (performance of the navigation and motion control and especially collision avoidance systems depend on meteorological factors). It is more convenient to denote the stated dependency for this case as $\theta(m(t, r), r)$, where $m(t, r)$ is a scalar deterministic function describing the “overall level” of meteorological conditions at a point with a time-location coordinate (t, r) . It is worth noting once more that a starting point has a coordinate $(0, 0)$.

Assume, for simplicity, as previously, a constant speed for the ship. Then the probability of an accident at a point on the route with coordinate (t, r) is $\theta(m(\frac{t}{v}, r), r)$, as $t = r/v$ is the time when the ship reaches this point. It is important that everything is again parameterized by only the variable r and equations (5)–(6) can be used for obtaining $P(\bar{R})$ – the probability of passing the route R_{ξ_1, ξ_2} without accident:

$$P(\bar{R}) = \exp \left\{ - \int_0^{\bar{R}} \theta \left(m \left(\frac{r}{v}, r \right), r \right) h_e(r) dr \right\}. \quad (23)$$

It is clear that for the practical application of this approach a model for the corresponding impact of the weather conditions on the probability of an accident should be developed. A simplified version, that can easily be generalized, is as follows. Consider 3 levels of weather severity: normal, severe and very severe. Let $\theta_0(r)$ be the baseline probability of an accident under normal weather conditions. Suppose that

$$0 < \theta_0(r) < \theta_1(r) \equiv k_1 \theta_0(r) < \theta_2(r) \equiv k_2 \theta_0(r) < 1; \quad \forall r \in [0, R] \quad (24)$$

where $1 < k_1 < k_2$ are constants to be defined for the concrete application. The number of levels can be easily increased. Therefore, using relation (24) for the given weather forecast and the scheduled route, one can derive $P(\bar{R})$ for this specific case. The weather forecast needed for this derivation should be presented in the following form: the level ($i = 0, 1, 2, \dots$) of weather conditions in a point with coordinate r at each instant of time $t \in [0, \bar{t}]$, where $\bar{t}$ is the horizon of the forecast. In this case the probability of an accident in the integrand in the right-hand side of (23), using relations (24), should be calculated as:

$$\theta\left(m\left(\frac{r}{v}, r\right), r\right) = k_i \theta_0(r); \quad k_0 \equiv 1$$

for the meteorological levels $i = 0, 1, 2, \dots$. This descriptive approach is illustrated by the following simplified example.

Example 2.

Suppose that navigational planning for the scheduled route R_{ξ_1, ξ_2} of length $\bar{R}$ (which takes into account the corresponding meteorological forecast and the scheduled speed of a ship) results in the following: for the first third of the route the meteorological conditions will be of level "0", for the second of level "2" and for the last one-of level "1". Then it is easy to see that the forecast probability of passing the route without accident is

$$\begin{aligned} P(\bar{R}) = & \exp \left\{ - \int_0^{\bar{R}/3} \theta_0(r) h_e(r) dr \right\} \\ & \times \exp \left\{ - \int_{\bar{R}/3}^{2\bar{R}/3} k_2 \theta_0(r) h_e(r) dr \right\} \\ & \times \exp \left\{ - \int_{2\bar{R}/3}^{\bar{R}} k_1 \theta_0(r) h_e(r) dr \right\}. \end{aligned}$$

The forecast of this kind (as every forecast) is uncertain to some extent, and therefore, the situation can also be modelled probabilistically. This means that the function $m(t, r)$ turns to a stochastic process and the corresponding failure rate $\lambda_a(r)$ turns to a hazard rate process. Some results of section 2 can be used for analyzing the latter case but this topic needs a more detailed study in the future.

4. Concluding remarks

This article generalizes results of Finkelstein (1998) to the cases of deterministic and stochastic external covariates. The suggested approach is based on the r -parameterization of the functions under consideration and subsequent reduction of the dimensionality of the model to the well-known one-dimensional case.

The stochastic component can be introduced directly in probability $\theta(t)$ and in the equivalent rate $h_e(r)$ as well. Both of these cases were considered in sections 2 and 3. It is shown that when the stochastic process is modelled by including the random parameter Y in the probability $\theta(t, Y)$, then the shape of the resulting (observed) failure rate of the time to an accident can differ dramatically from the corresponding failure rate for a fixed (realized) value of Y .

The method of this paper develops a basis for obtaining optimal solutions. The following problem, for instance, can be approached using it: define a route connecting ξ_1 and ξ_2 with a maximal probability $P(\bar{R})$ with or without specific restrictions (time on the route, fuel consumption etc.).

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